

Tanish Dalal

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EDUCATION

University of Michigan

B.S.E. in Computer Engineering | **GPA:** 3.83/4.00

Ann Arbor, MI
Aug. 2024 - May 2028

Honors: Dean's List, University Honors | **Societies:** Tau Beta Pi Engineering Honor Society

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Architecture, Embedded Systems, Discrete Mathematics, Linear Algebra, Logic Design

EXPERIENCE

Software Engineering Intern

Candela Technologies

Jul. 2026 – Present
Ferndale, WA (Remote)

- Integrated 2D LiDAR and IMU sensor data with ROS 2 Cartographer using Ansible, tuning localization and mapping parameters to support autonomous site surveying across **6** clients' test environments.
- Built multithreaded Python automation pipelines interacting with LANforge REST APIs across **15** access points, reducing network test turnaround from **3 days** to **4 hours** while eliminating **95%** of manual data-entry errors.
- Synchronized real-time robot telemetry with wireless network metrics (RSSI, latency) using hardware time synchronization via ptp4l/phc2sys, automating the mapping of network performance to spatial coordinates.
- Resolved ROS 2 connectivity bottlenecks across multi-subnet LAN environments by optimizing CycloneDDS discovery configurations and analyzing Wireshark packet captures, reducing testing downtime by **30%**.

Software Engineering Intern

ADNOC Distribution, EVCI & H2 Department

May 2025 – Jul. 2025
Abu Dhabi, UAE

- Built an OCPP 1.6 telemetry ingestion service in Python/FastAPI for WebSocket connections from **400+** EV chargers, normalizing **MeterValues** and **StatusNotification** events into TimescaleDB at **7k+** points/min with **<100 ms** p99 end-to-end latency.
- Engineered an XGBoost predictive-maintenance model using **40+** rolling-window features derived from EV-charger voltage, current, thermistor, and session telemetry, achieving **0.78 PR-AUC** and detecting hardware failures a median of 30 min before shutdown.
- Built automated regression tests with **pytest** for OCPP message handling, covering malformed payloads, charger reconnections, duplicate events, and database write failures, reducing manual validation time by **60%**.

PROJECTS

Computer Vision Lead & Co-Founder — Michigan Robomasters

C++, Python, ROS 2, OpenCV, RViz, STM32

Aug. 2025 – Present

- Lead a **6-member** team developing a ROS 2 auto-aiming system; trained YOLO26n on 8k+ labeled armor-plate frames to **0.58 mAP@50:95** and integrated **solvePnP** for 3D pose estimation, achieving **<1.5° angular error** at **45 FPS** on a Jetson Orin Nano.
- Implemented a constant-velocity Kalman filter for enemy armor-plate tracking and projectile lead compensation, reducing angular tracking error by **28%** during autonomous shooting tests.
- Developed autonomous navigation for the sentry robot in ROS 2, integrating FAST-LIO2 SLAM, Livox MID-360 LiDAR, and Nav2; achieved a **90%** collision-free route-completion rate across dynamic multi-waypoint tests in Gazebo and on the physical robot.

Algorithm Practice Platform

React, FastAPI, Python, JavaScript, SQLite

Sep. 2025 – Present

- Built a real-time multiplayer algorithm battle platform using FastAPI and WebSockets, supporting **50+ concurrent matches** with **<150 ms** median latency for **200+** students.
- Engineered a secure code execution engine supporting 3 languages, integrating a C++ assembly viewer (*-O0* vs. *-O2*), log-log regression Big-O analysis, and TF-IDF + k-means algorithmic approach classification.
- Implemented ELO-seeded matchmaking, 8-player tournament brackets with **<2s** queues, and post-match rating updates via SQLite.
- Built an AI problem and interview generator for automated coding assessments, achieving a **96% validation success rate** with **<1.2s** response times.

Robotic Arm Controls Developer — Michigan Mars Rover

C++, Python, ROS 2, RViz, Eigen, NumPy

Aug. 2024 – Present

- Engineered a closed-loop Cartesian IK controller for a **5-DOF** robotic arm, improving target alignment accuracy by **20%** using proportional velocity control and shortest-path end-effector tracking.
- Developed geometric IK solvers integrated into a **30 Hz** ROS 2 velocity pipeline with joint-limit checks, achieving **<1 cm** convergence and reducing motor saturation by **30%**.
- Built a Python workspace point-cloud visualizer sampling reachability at **5 cm** resolution with dexterity scoring and draggable RViz obstacles, cutting manual validation time by **50%**.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, SQL, Java, Assembly, Verilog, MATLAB

ML & Vision: PyTorch, Ultralytics YOLO, OpenCV, XGBoost, scikit-learn, pandas, NumPy

Robotics & Embedded: ROS 2, Nav2, SLAM (Cartographer, FAST-LIO2), RViz, Gazebo, Eigen, CycloneDDS

Frameworks & Data: FastAPI, React, WebSockets, REST APIs, PostgreSQL, TimescaleDB, SQLite

Tools & Systems: Linux, Git, GitHub Actions, Ansible, Wireshark, pytest, ModelSim

Embedded: NVIDIA Jetson Orin, STM32